▧ DOCUMENT 1 — Detailed Summation of the Conversation

Overview

You and I have been constructing a practical, experimentally grounded workflow for probing, mapping, and modifying the behavioral topology of a language model. The core idea is that a model's responses to repeated sampling on the same prompts form clusters, minor basins, and runners, which together reveal the model's internal "shadow geometry."

Your goal is to:

Obtain prompts and multi-trajectory responses from a single model.

Embed and cluster those responses to reconstruct the topology.

Run the same model locally to lightly re-probe the prompts and confirm the topology.

Perform topology surgery by collapsing or removing a minor cluster.

Re-probe to confirm the topology has changed.

This is a closed-loop experiment in model behavior modification.

Key Points From the Conversation
1. You need multi-trajectory data
You want a dataset where:

A single model

Answered many prompts

Multiple times each (10+ samples per prompt)

This allows you to map the model’s attractor basins.

2. You do not need to generate the full dataset yourself
Your hardware cannot generate thousands of trajectories for a large model.
But you don’t need to — you only need:

The prompts

The recorded trajectories

The ability to run the same model locally

3. You can embed the recorded responses
Embedding the responses lets you:

Cluster them

Identify major and minor basins

Detect runners

Reconstruct the topology

This is the “shadow map.”

4. You lightly re-probe the model locally
You only need:

5–10 samples per prompt

Same model checkpoint

Same sampling settings

If your local clusters match the dataset clusters, the topology is confirmed.

5. You perform topology surgery
You want to remove a minor cluster.
This can be done via:

A small fine-tune (LoRA)

A steering vector

A local router/patch

Logit biasing

The goal is to collapse the minor basin.

6. You confirm the topology has changed
After the edit:

Re-probe

Re-embed

Re-cluster

Compare before/after

If the minor cluster disappears or shrinks dramatically, the surgery succeeded.

The Core Workflow (Compressed)
Acquire prompts + multi-trajectory responses.

Embed and cluster to map topology.

Download the same model checkpoint.

Lightly re-probe prompts locally.

Confirm topology matches.

Identify a minor cluster.

Perform topology surgery.

Re-probe.

Re-cluster.

Confirm the minor basin collapsed.

This is the full experimental loop.

▨ DOCUMENT 2 — Start of the Paper
Below is a clean, formal beginning of your paper, containing:

Statement of the problem

Detailed hypothesis

Detailed thesis

Structure that leaves room for the eventual conclusion

You can expand this into a full paper later.

Title (Placeholder)
Topology-Preserving and Topology-Modifying Interventions in Language Model Behavior

1. Statement of the Problem
Large language models exhibit complex, multi-modal behavioral patterns when repeatedly sampled on identical prompts. These patterns are not random noise; they form stable clusters, minor attractor basins, and rare runner trajectories that together constitute a model’s behavioral topology.

However, this topology is poorly understood, rarely measured, and almost never manipulated

directly.
The central problem is:

How can we map the behavioral topology of a language model, confirm its stability, and then deliberately modify it by removing or collapsing specific minor attractor basins?

This paper proposes a practical, experimentally grounded method for doing exactly that.

2. Hypothesis (Detailed)

2.1 Core Hypothesis

A language model’s repeated responses to the same prompt form a stable, reproducible topology in embedding space. This topology consists of:

Major attractor clusters

Minor attractor clusters

Runner trajectories

Curvature and basin boundaries

This topology is:

Model-specific

Stable under repeated sampling

Reproducible across hardware

Sensitive to fine-tuning, quantization, and pruning

Modifiable through targeted interventions

2.2 Sub-Hypotheses

Topology Stability Hypothesis

The cluster structure of a model’s responses remains consistent across repeated sampling and across different machines running the same checkpoint.

Topology Identifiability Hypothesis

Embedding-space clustering of multi-trajectory responses is sufficient to reconstruct the model’s behavioral topology.

Topology Surgery Hypothesis

It is possible to deliberately collapse or

remove a minor attractor basin through targeted fine-tuning, steering, or local adapters.

Topology Drift Hypothesis
After intervention, the model’s topology will measurably change, and this change can be confirmed through re-sampling and re-clustering.

3. Thesis (Detailed)

3.1 Thesis Statement

This paper argues that the behavioral topology of a language model is a measurable, manipulable object. By combining multi-trajectory datasets, embedding-space clustering, and targeted model interventions, we can:

Map the topology

Confirm its stability

Perform controlled modifications

Verify the resulting topological changes

This establishes a new experimental paradigm for understanding and shaping model behavior.

3.2 Thesis Components

A. Topology as a Behavioral Object

The model’s “mind” is not its weights but the geometry of its responses under repeated sampling.

B. Topology Reconstruction

Embedding and clustering of multi-trajectory responses reveal the attractor structure.

C. Topology Confirmation

Light local re-sampling confirms that the topology is stable and reproducible.

D. Topology Modification

Minor attractor basins can be collapsed through:

fine-tuning

steering vectors

local adapters

logit biasing

E. Topology Verification
Re-sampling and re-clustering confirm that the basin has been removed or altered.

4. Structure of the Paper
4.1 Introduction
Motivation

Background

Prior work

Why topology matters

4.2 Methods
Dataset acquisition

Embedding and clustering

Local re-probing

Intervention techniques

4.3 Experiments
Mapping topology

Identifying minor clusters

Performing topology surgery

Measuring topology drift

4.4 Results
Before/after cluster maps

Stability metrics

Drift metrics

4.5 Discussion
Implications

Limitations

Future directions

4.6 Conclusion (to be written later)
Reserved for final synthesis once experiments are complete.

If you want, I can expand either document, turn the second into a full paper, or help you

design the experimental section next.